

NeoCortex — Technical Summary

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Overview

NeoCortex is an open-source spiking-network simulator that runs a 48-hour virtual drug exposure in under a minute on a single GPU. It models 100,000 leaky integrate-and-fire neurons and about 10 million synapses, with six neuromodulatory systems and four plasticity mechanisms.

A drug enters as a change to a transporter, enzyme, receptor or channel, with its own pharmacokinetics. The simulator then reports neurotransmitter levels and eight physiological readouts over time.

NeoCortex is a research prototype. We are looking for a partner to validate it prospectively and to help integrate it into an existing in silico pipeline.

Architecture

The model has four layers, each feeding the next at every 1 ms step.

Layer	Components	Role
Neural core	100k LIF neurons, 4 cortical columns × 6 layers, pyramidal and PV/SST/VIP/chandelier interneurons	Spike generation and propagation
Neuromodulation	5-HT, DA, NE, ACh, GABA, glutamate; 13 receptor subtypes	Release, reuptake and decay dynamics; Hill-equation receptor activation
Plasticity	BCM, synaptic scaling, dopamine-gated three-factor learning, short-term plasticity	Keeps the network stable under chronic perturbation
Physiology	8 readouts (fatigue, muscle tension, pain, heart rate and others)	Weighted mapping from neurotransmitter state, 48 literature-based coefficients

Pharmacology. Each drug has a one-compartment PK profile and Hill-equation target occupancy, with coefficients of 1.2–1.8. Ten drugs include an active metabolite with its own kinetics, such as norfluoxetine ($t_{1/2}$ 336 h).

Disease models. Each disease is defined by neurotransmitter efficacy deficits. For example, the MDD model sets 5-HT at –40%, DA at –30% and NE at –25%.

Stack. PyTorch with sparse CSR connectivity on GPU. An Intel Loihi port is included, with a NumPy fallback when no board is present. Full methods and results: github.com/inkman97/neocortex (paper NeoCortex_Final_v6.pdf).

Performance

On one NVIDIA A100, a full 48-hour virtual trial takes 45–60 seconds. That covers 2,000 warm-up steps plus 2,880 treatment steps at 10–15 ms each.

Network size	Time per step
100k neurons	10–15 ms
500k neurons	42 ms
1M neurons	95 ms

A genetic-algorithm screen of 20 candidate profiles takes about 40 seconds when batched on GPU. Multi-GPU scaling is in development.

Retrospective results

In three retrospective case studies, the direction of the simulated 48-hour effect matched the known clinical outcome. These are qualitative, directional agreements, not quantitative predictions of clinical endpoints.

Case	Disease model	48 h simulated change	Known clinical outcome	Agreement
Fluoxetine 20 mg	MDD	5-HT 29.1% → 43.0%; fatigue 60% → 51%	Approved; effective	Direction matches

Case	Disease model	48 h simulated change	Known clinical outcome	Agreement
Reboxetine 8 mg	MDD	NE 22% → 25%; 5-HT 27.7% → 28.4%; fatigue 61% → 55%	Not approved by FDA; efficacy not shown in meta-analysis (Eyding et al., BMJ 2010)	Direction matches
Lamotrigine 200 mg	Epilepsy (high Glu, low GABA)	E/I ratio 0.29 → 0.10; Glu -18%; GABA +19%	Effective for seizure reduction	Mechanism direction matches
Levodopa 100 mg	Advanced Parkinson's (DA 5.7%)	DA 5.7% → 6.0%; fatigue 56% → 50%	Limited response in advanced disease	Direction matches

Three caveats apply. The physiology coefficients were set from the literature, so serotonin's weight in mood-related readouts is a modelling choice, not an emergent finding. Runs are stochastic and figures come from single runs. Clinical endpoints are measured at 4–8 weeks, while these simulations cover 48 hours.

Limitations

The main gap is prospective validation: no blinded test has been run yet. The other known limits are listed below.

- **Time horizon:** validated only at 48 hours. The engine accepts runs of up to one year, but longer windows have not been checked against data.
- **Reproducibility:** no fixed random seed yet. Multi-seed runs with confidence intervals are the next release.
- **Coverage:** three pathologies and about 20 drugs. No CYP450 drug–drug interactions and no pharmacogenomic variants.
- **Physiology mapping:** the 48 coefficients are literature-derived and not yet fitted to patient-level data.
- **Drug discovery module:** it optimises target-potency profiles. The example molecule (a dual AChE/SERT inhibitor) has no binding or ADME data and should be read as a hypothesis.

Proposed collaboration

We propose to start with a free blind test. If it succeeds, NeoCortex could become a mechanistic CNS module within an existing in silico platform.

1. **Blind test (2–3 weeks, no cost).** The partner chooses 3–5 CNS compounds with known outcomes and does not disclose them. We submit predictions, with multi-seed confidence intervals, before the outcomes are revealed.
2. **Pilot integration (about 3 months).** NeoCortex runs as a mechanistic CNS model alongside the partner's QSP and PK/PD models. Its readouts feed the partner's virtual-population and trial-design tools.
3. **Long-term model.** Options are a commercial licence, joint development, or co-authored validation work. All of these are compatible with the open-source core.

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